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Ibrahim Odat

AI Security Researcher · LLM-Enabled Autonomous Systems Security

ibrahimodat@oakland.edu | ibrahimodatt@gmail.com | Michigan, USA
3odat.github.io | linkedin.com/in/ibrahim-odat | Google Scholar | ORCID 0009-0009-3175-0473

SUMMARY

AI security researcher specializing in the security of large language model (LLM)-enabled autonomous drone systems and multi-agent unmanned aerial vehicle (UAV) architectures. Holds an M.S. in Cybersecurity (GPA 4.0/4.0) from Oakland University and is pursuing a Ph.D. in Computer Science & Informatics under Dr. Anyi Liu, with research addressing emerging vulnerabilities in AI-driven autonomous systems with implications for national defense, border surveillance, and critical infrastructure protection. Author of three peer-reviewed IEEE and IEEE/ACM publications and one preprint, cited 52 times (Google Scholar, November 2026). Current doctoral research identifies memory as a control-plane attack surface in LLM-based multi-agent systems, with a defense pipeline that reduces measured attack success from 100% to 0% in controlled evaluation.

EDUCATION

Oakland University — Rochester, MI
Ph.D. Candidate, Computer Science & Informatics · GPA 3.93/4.0 — Jan. 2025 – Expected Jan. 2029

- Dissertation (in progress): Security of LLM-Enabled Multi-Agent UAV Systems
- Advisor: Dr. Anyi Liu, Department of Computer Science & Engineering
- Research focus: adversarial threats, memory-poisoning attacks, and defense mechanisms for LLM-powered autonomous drone and multi-agent systems

Oakland University — Rochester, MI
M.S., Cybersecurity · GPA 4.0/4.0 — May 2022 – Dec. 2023

PUBLICATIONS

Total citations: 52 (Google Scholar, November 2026)

When Memory Hurts: Amplifying Mission-Channel Attacks in LLM-Driven UAV Agents — 2026
25th International Conference on Machine Learning and Applications (ICMLA 2026), pp. 693–700

- Authors: **Odat, I.**, Tawayha, H., & Liu, A. (First Author) · doi:10.1109/ICMLA69890.2026.00101
- Measured whether bounded short-term mission memory amplifies one-shot corruptions of the evidence a UAV agent sees, across four attack pathways, 12 planners, and 1,920 runs in a PX4/MAVSDK-compatible mission framework
- Showed that bounded memory raises attack-side mission failure from 31.5% to 53.3%, with false perception amplified 2.83×; a write-time Memory Contract Monitor reduced belief persistence to zero under direct memory poisoning

Net-GPT: A LLM-Empowered Man-in-the-Middle Chatbot for Unmanned Aerial Vehicle — 2023
IEEE/ACM Symposium on Edge Computing (SEC 2023), pp. 287–293

- Authors: Piggott, B., Patil, S., Feng, G., **Odat, I.**, Mukherjee, R., Dharmalingam, B., & Liu, A. · doi:10.1145/3583740.3626809
- Contributed core LLM framework design, system implementation, and experimental evaluation of the attack pipeline; developed and validated the LLM fine-tuning methodology that achieved 95.3% accuracy in generating counterfeit UAV network packets, enabling automated man-in-the-middle (MITM) attacks on drone-to-ground-station communications
- Cited 50 times on Google Scholar as of November 2026, by researchers in robotic systems, smart grid security, and flying communication networks

Heterogeneous Generative Dataset for UASes 2023
IEEE International Conference on Mobility, Operations, Services and Technologies (MOST 2023), pp. 229–230

- Authors: Dharmalingam, B., **Odat, I.**, Mukherjee, R., Piggott, B., & Liu, A. (Second Author) · doi:10.1109/MOST57249.2023.00034
- Contributed to dataset design and evaluation methodology for standardized benchmarking of adversarial attacks on unmanned aerial systems; addresses the absence of heterogeneous training data for UAS security evaluation

PREPRINTS & MANUSCRIPTS UNDER REVIEW

AeroMind: Poisoning the Control Plane of LLM-Driven UAV Agents 2026
Under review at ACM AsiaCCS 2027

- Authors: **Odat, I.**, Liu, A., & Li, Y. (First Author)
- Evaluated a multi-agent UAV system across 15 attack scenarios and seven planner backends, demonstrating that three poisoned memory records can redirect coordinated mission behavior and spread between agents through a shared-memory architecture
- Developed a three-layer defense pipeline (provenance, trust reranking, diversity cap) reducing measured attack success from 100% to 0% in controlled evaluation
- Proposed and formalized the first documented threat model for memory-poisoning attacks in LLM-powered multi-agent UAV systems

Beyond Belief: Evaluating and Mitigating LLM Susceptibility to Misinformation 2026
arXiv preprint arXiv:2610.33962

- Authors: Kanaan, E., Lee, J., **Odat, I.**, Vosoughi, S., & Ma, W.
- Tests whether the credibility judgments of tool-using LLMs predict whether they act on misinformation, across five models, 196 constructed false notices, and 204 archived attacks, with forewarning and fine-tuning as mitigations

RESEARCH EXPERIENCE

Graduate Research Assistant (GRA) Jan. 2022 – Present
Oakland University · Advisor: Dr. Anyi Liu Rochester, MI

- Designed and implemented AeroMind, a multi-agent UAV fleet integrating large language models (LLMs), persistent vector memory, and PX4 software-in-the-loop (SITL) flight simulation; serves as the primary research platform for the doctoral dissertation on memory-poisoning attacks and defenses in autonomous UAV systems
- Penetration-tested PX4 open-source drone flight controllers, identifying buffer overflows, session hijacking, and MAVLink (drone communication protocol) exploits across three distinct attack vectors; developed and validated mitigations incorporated into the lab's standardized UAV security evaluation framework
- Architected AI-driven threat detection pipelines using fine-tuned LLMs, prompt engineering, and Ollama, demonstrating a 60% reduction in manual threat triage time in controlled lab evaluation
- Simulated MITM and brute-force attacks against UAV encryption protocols and IDS/IPS configurations; findings documented in technical reports and incorporated into subsequent UAV security research within the lab

TEACHING EXPERIENCE

Teaching Assistant (TA) Jan. 2022 – Dec. 2024
Oakland University, Department of Computer Science & Engineering Rochester, MI

- Teaching Assistant for four graduate and undergraduate cybersecurity and systems courses, supporting instruction for both M.S. and B.S. level students
- **CSI 5480: Information Security Practices** · graduate-level instruction on security principles, threat modeling, and defensive techniques
- **CSI 6600: Network Security** · graduate-level coursework covering network threat analysis, intrusion detection, and secure communications
- **CSI 4500/5500: Fundamentals of Operating Systems** · OS architecture, process management, and system-level security concepts
- **CSI 1420: Introduction to C Programming and Unix** · undergraduate systems programming fundamentals

PRESENTATIONS & RESEARCH OUTREACH

Research Presentation, Engineering Day Jan. 24, 2026
Oakland University School of Engineering and Computer Science Rochester, MI

- Presented doctoral research on LLM-enabled autonomous drone security to prospective students, families, and faculty at the annual Engineering Day public outreach event
- Demonstrated the AeroMind multi-agent drone framework end to end, covering the autonomy pipeline, real-time control, AI perception and decision-making, and memory-based attacks and defenses across DJI Tello and PX4-based drone platforms

OPEN-SOURCE RESEARCH PROJECTS

AeroMind Drone Security Testbed github.com/OdatSec/AeroMind

- Multi-agent UAV research platform with a supervisor agent, autonomous scout drones, and persistent shared memory, built to evaluate memory poisoning, cross-agent propagation, and retrieval-stage defenses. Python, PX4 SITL, MAVSDK, Gazebo, Ollama

Net-GPT Evaluation Toolkit github.com/OdatSec/Net-GPT

- Reusable evaluation workflows for fine-tuning language models and generating counterfeit UAV network packets, released to enable independent replication of the SEC 2023 paper. Python, LLM fine-tuning, MAVLink

PX4 Autopilot Session Hijacking Exploit PX4, MAVLink, MAVSDK, Python

- Reverse-engineered MAVLink session management to achieve privilege escalation on PX4 Autopilot; built a proof-of-concept exploit demonstrating telemetry stream hijacking, with remediation proposals documented for the research community

TECHNICAL SKILLS

AI/LLM Security: Adversarial ML, LLM fine-tuning (QLoRA), prompt engineering, RAG poisoning, memory poisoning, LangGraph, LangChain, Ollama

Autonomous Systems: ROS 2, PX4 Autopilot, Gazebo, MAVSDK, MAVLink, NVIDIA Jetson Nano, YOLOv8

Offensive Security: Penetration testing, exploit development, firmware analysis, reverse engineering, red teaming, OSINT

Frameworks & Standards: MITRE ATT&CK, OWASP Top 10, NIST SP 800-171, NIST AI RMF, PCI DSS, ISO 27001

Security Tooling: Kali Linux, Metasploit, Burp Suite, Wireshark, Nessus, Nmap, SQLMap, SIEM, IDS/IPS

Programming & Infrastructure: Python, Bash, C/C++, SQL, FastAPI, Docker, AWS

CERTIFICATIONS & MEMBERSHIPS

Certified Ethical Hacker (CEH), EC-Council, 2023 · **eLearnSecurity Junior Penetration Tester (eJPT)**, INE, 2022 · **Fortinet NSE 1 & NSE 2**, Fortinet, 2021

Member, **IEEE** · Member, **ACM**

PRIOR INDUSTRY EXPERIENCE

Cybersecurity Specialist Jan. 2021 – Jan. 2022
ZADD Amman, Jordan

- Integrated AI-based anomaly detection into an enterprise SaaS platform, reducing false-positive security alerts by 35%; managed enterprise-wide NIST SP 800-171 compliance remediation across critical infrastructure

Penetration Tester Jan. 2020 – Jan. 2021
Futuretec Security Solutions Amman, Jordan

- Conducted systematic web application and API security assessments identifying critical injection, authentication, and configuration vulnerabilities; developed custom Python automation frameworks reducing manual testing effort by 50%